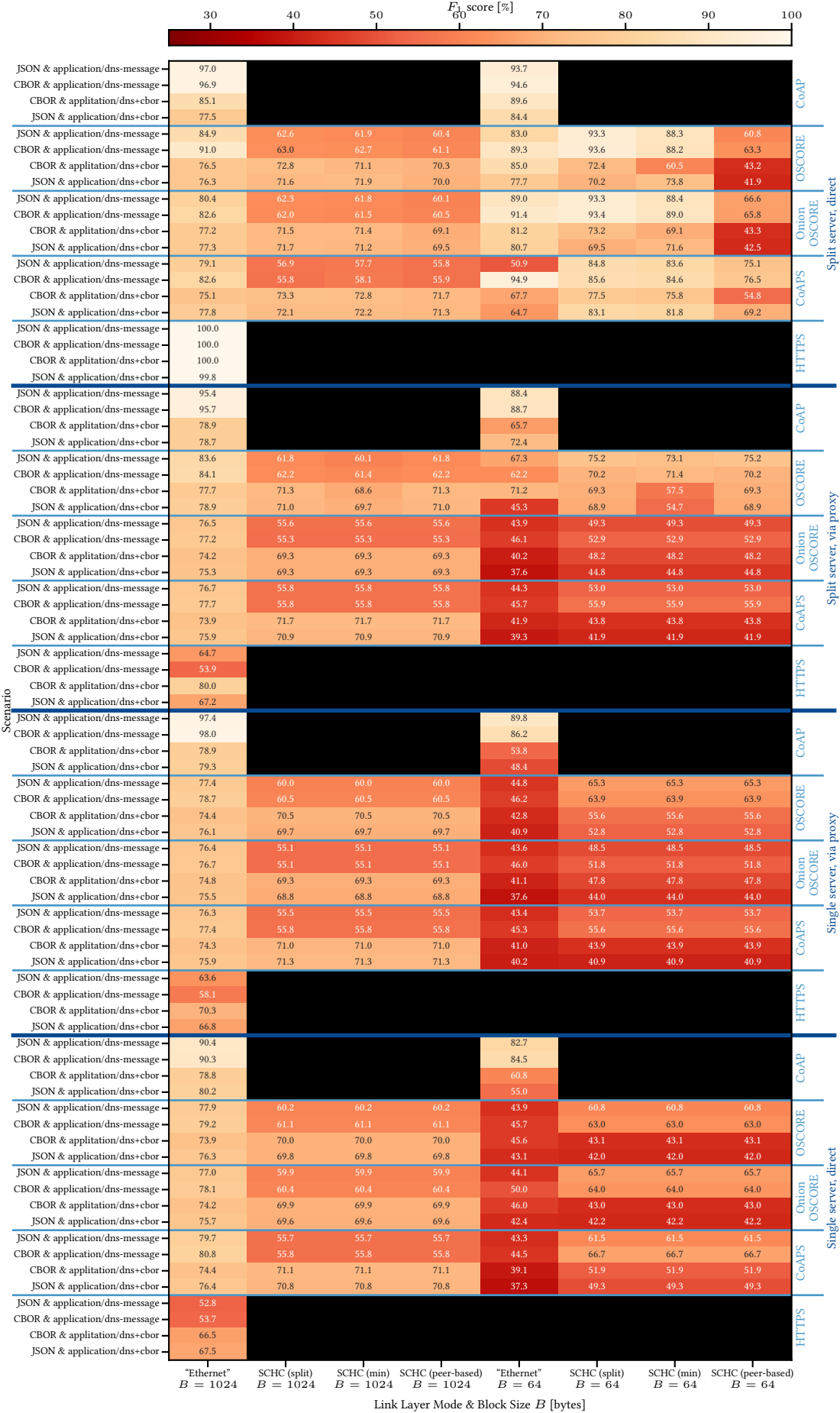
F_1 score [%]
30
40
50
60
70
80
90
100
Scenario
Link Layer Mode & Block Size B [bytes]
"Ethernet" $B = 1024$
SCHC (split) $B = 1024$
SCHC (min) $B = 1024$
SCHC (peer-based) $B = 1024$
"Ethernet" $B = 64$
SCHC (split) $B = 64$
SCHC (min) $B = 64$
SCHC (peer-based) $B = 64$
Split server, direct
JSON & application/dns-message 97.0 93.7
CBOR & application/dns-message 96.9 94.6
CBOR & applitation/dns+cbor 85.1 89.6
JSON & application/dns+cbor 77.5 84.4
CoAP
JSON & application/dns-message 84.9 62.6 61.9 60.4 83.0 93.3 88.3 60.8
CBOR & application/dns-message 91.0 63.0 62.7 61.1 89.3 93.6 88.2 63.3
CBOR & applitation/dns+cbor 76.5 72.8 71.1 70.3 85.0 72.4 60.5 43.2
JSON & application/dns+cbor 76.3 71.6 71.9 70.0 77.7 70.2 73.8 41.9
OSCORE
JSON & application/dns-message 80.4 62.3 61.8 60.1 89.0 93.3 88.4 66.6
CBOR & application/dns-message 82.6 62.0 61.5 60.5 91.4 93.4 89.0 65.8
CBOR & applitation/dns+cbor 77.2 71.5 71.4 69.1 81.2 73.2 69.1 43.3
JSON & application/dns+cbor 77.3 71.7 71.2 69.5 80.7 69.5 71.6 42.5
Onion OSCORE
JSON & application/dns-message 79.1 56.9 57.7 55.8 50.9 84.8 83.6 75.1
CBOR & application/dns-message 82.6 55.8 58.1 55.9 94.9 85.6 84.6 76.5
CBOR & applitation/dns+cbor 75.1 73.3 72.8 71.7 67.7 77.5 75.8 54.8
JSON & application/dns+cbor 77.8 72.1 72.2 71.3 64.7 83.1 81.8 69.2
CoAPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 99.8
HTTPS
Split server, via proxy
JSON & application/dns-message 95.4 88.4
CBOR & application/dns-message 95.7 88.7
CBOR & applitation/dns+cbor 78.9 65.7
JSON & application/dns+cbor 78.7 72.4
CoAP
JSON & application/dns-message 83.6 61.8 60.1 61.8 67.3 75.2 73.1 75.2
CBOR & application/dns-message 84.1 62.2 61.4 62.2 62.2 70.2 71.4 70.2
CBOR & applitation/dns+cbor 77.7 71.3 68.6 71.3 71.2 69.3 57.5 69.3
JSON & application/dns+cbor 78.9 71.0 69.7 71.0 45.3 68.9 54.7 68.9
OSCORE
JSON & application/dns-message 76.5 55.6 55.6 55.6 43.9 49.3 49.3 49.3
CBOR & application/dns-message 77.2 55.3 55.3 55.3 46.1 52.9 52.9 52.9
CBOR & applitation/dns+cbor 74.2 69.3 69.3 69.3 40.2 48.2 48.2 48.2
JSON & application/dns+cbor 75.3 69.3 69.3 69.3 37.6 44.8 44.8 44.8
Onion OSCORE
JSON & application/dns-message 76.7 55.8 55.8 55.8 44.3 53.0 53.0 53.0
CBOR & application/dns-message 77.7 55.8 55.8 55.8 45.7 55.9 55.9 55.9
CBOR & applitation/dns+cbor 73.9 71.7 71.7 71.7 41.9 43.8 43.8 43.8
JSON & application/dns+cbor 75.9 70.9 70.9 70.9 39.3 41.9 41.9 41.9
CoAPS
JSON & application/dns-message 64.7
CBOR & application/dns-message 53.9
CBOR & applitation/dns+cbor 80.0
JSON & application/dns+cbor 67.2
HTTPS
Single server, via proxy
JSON & application/dns-message 97.4 89.8
CBOR & application/dns-message 98.0 86.2
CBOR & applitation/dns+cbor 78.9 53.8
JSON & application/dns+cbor 79.3 48.4
CoAP
JSON & application/dns-message 77.4 60.0 60.0 60.0 44.8 65.3 65.3 65.3
CBOR & application/dns-message 78.7 60.5 60.5 60.5 46.2 63.9 63.9 63.9
CBOR & applitation/dns+cbor 74.4 70.5 70.5 70.5 42.8 55.6 55.6 55.6
JSON & application/dns+cbor 76.1 69.7 69.7 69.7 40.9 52.8 52.8 52.8
OSCORE
JSON & application/dns-message 76.4 55.1 55.1 55.1 43.6 48.5 48.5 48.5
CBOR & application/dns-message 76.7 55.1 55.1 55.1 46.0 51.8 51.8 51.8
CBOR & applitation/dns+cbor 74.8 69.3 69.3 69.3 41.1 47.8 47.8 47.8
JSON & application/dns+cbor 75.5 68.8 68.8 68.8 37.6 44.0 44.0 44.0
Onion OSCORE
JSON & application/dns-message 76.3 55.5 55.5 55.5 43.4 53.7 53.7 53.7
CBOR & application/dns-message 77.4 55.8 55.8 55.8 45.3 55.6 55.6 55.6
CBOR & applitation/dns+cbor 74.3 71.0 71.0 71.0 41.0 43.9 43.9 43.9
JSON & application/dns+cbor 75.9 71.3 71.3 71.3 40.2 40.9 40.9 40.9
CoAPS
JSON & application/dns-message 63.6
CBOR & application/dns-message 58.1
CBOR & applitation/dns+cbor 70.3
JSON & application/dns+cbor 66.8
HTTPS
Single server, direct
JSON & application/dns-message 90.4 82.7
CBOR & application/dns-message 90.3 84.5
CBOR & applitation/dns+cbor 78.8 60.8
JSON & application/dns+cbor 80.2 55.0
CoAP
JSON & application/dns-message 77.9 60.2 60.2 60.2 43.9 60.8 60.8 60.8
CBOR & application/dns-message 79.2 61.1 61.1 61.1 45.7 63.0 63.0 63.0
CBOR & applitation/dns+cbor 73.9 70.0 70.0 70.0 45.6 43.1 43.1 43.1
JSON & application/dns+cbor 76.3 69.8 69.8 69.8 43.1 42.0 42.0 42.0
OSCORE
JSON & application/dns-message 77.0 59.9 59.9 59.9 44.1 65.7 65.7 65.7
CBOR & application/dns-message 78.1 60.4 60.4 60.4 50.0 64.0 64.0 64.0
CBOR & applitation/dns+cbor 74.2 69.9 69.9 69.9 46.0 43.0 43.0 43.0
JSON & application/dns+cbor 75.7 69.6 69.6 69.6 42.4 42.2 42.2 42.2
Onion OSCORE
JSON & application/dns-message 79.7 55.7 55.7 55.7 43.3 61.5 61.5 61.5
CBOR & application/dns-message 80.8 55.8 55.8 55.8 44.5 66.7 66.7 66.7
CBOR & applitation/dns+cbor 74.4 71.1 71.1 71.1 39.1 51.9 51.9 51.9
JSON & application/dns+cbor 76.4 70.8 70.8 70.8 37.3 49.3 49.3 49.3
CoAPS
JSON & application/dns-message 52.8
CBOR & application/dns-message 53.7
CBOR & applitation/dns+cbor 66.5
JSON & application/dns+cbor 67.5
HTTPS